

Lead Management System

Enterprise Lead Management System

Tech Stack

- React.js + Vite
- Node.js + Express.js
- Prisma ORM
- PostgreSQL
- JWT Authentication
- Socket.io
- Docker

Developed By

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Features

- JWT Authentication
- RBAC
- Lead Auto Assignment
- Activity Logs
- Realtime Notifications
- Dark/Light Mode Dashboard

PROJECT ARCHITECTURE

Project Architecture

Frontend Layer

React.js application responsible for:

- UI rendering
- API integration
- state management
- role-based routing
- dashboard analytics

Backend Layer

Node.js + Express.js server responsible for:

- REST APIs
- authentication
- business logic
- RBAC authorization
- lead assignment logic
- websocket communication

Database Layer

PostgreSQL + Prisma ORM responsible for:

- relational data management
- indexing
- transactions
- audit storage
- optimized queries

Realtime Communication

Socket.io used for:

- live lead assignment notifications
- realtime dashboard updates
- status change alerts

FOLDER STRUCTURE

Folder Structure

Backend Structure

backend/

- config/
- controllers/
- services/
- repositories/
- middlewares/
- routes/
- validations/
- utils/
- sockets/
- tests/
- prisma/
- logs/

Frontend Structure

frontend/

- api/
- components/
- pages/

- hooks/
- layouts/
- context/
- routes/
- validations/
- theme/

Benefits

- scalable architecture
- separation of concerns
- reusable components
- maintainability
- easier debugging

AUTHENTICATION FLOW

JWT Authentication Flow

Login Process

1. User enters credentials
2. Backend validates user
3. Password compared using bcrypt
4. Access token generated
5. Refresh token generated
6. Refresh token stored in DB
7. Access token returned to frontend

Security Features

- bcrypt password hashing
- httpOnly refresh cookies
- refresh token rotation
- JWT middleware

- RBAC authorization
- protected routes

Role-Based Access

ADMIN:

- full access

MANAGER:

- lead management access

AGENT:

- assigned leads only

DATABASE DESIGN

Database Design Decisions

Main Tables

users

Stores:

- admin
- managers
- agents

leads

Stores:

- customer lead data
- assignment details

- status tracking
- priority levels

activity_logs

Tracks:

- lead actions
- login actions
- assignment history

lead_status_history

Maintains:

- complete status transition audit trail

refresh_tokens

Handles:

- secure JWT token rotation

Design Decisions

- UUID primary keys
- indexed columns
- normalized schema
- foreign key constraints
- soft delete implementation

LEAD ASSIGNMENT LOGIC

Lead Auto Assignment Logic

Assignment Strategy

Least-Loaded Agent Assignment

Workflow

1. Manager/Admin creates lead
2. System fetches active agents
3. Count active open leads per agent
4. Find least-loaded agent
5. Assign lead automatically
6. Generate notification
7. Store activity log

Concurrency Handling

Used:

- Prisma transactions
- Serializable isolation level

Benefits

- balanced workload
- scalable architecture
- fair distribution
- future extensibility

SCALABILITY & SECURITY

Scalability & Security

Scalability Considerations

- modular backend architecture
- reusable services
- API versioning
- repository pattern

- Prisma ORM optimization
- websocket-ready design

Security Features

- JWT authentication
- RBAC middleware
- password hashing
- rate limiting
- helmet security
- validation middleware
- secure cookies
- audit logging

Deployment Ready

- Docker support
- environment configuration
- Swagger API docs

CHALLENGES FACED

Challenges Faced

Technical Challenges

- implementing secure JWT rotation
- handling role-based authorization
- implementing least-loaded assignment
- Prisma UUID validation issues
- realtime websocket synchronization
- maintaining clean architecture

Solutions Implemented

- centralized RBAC middleware
- UUID validation middleware

- Prisma transactions
- reusable API response handlers
- socket event abstraction
- modular folder structure

FUTURE IMPROVEMENTS

Future Improvements

Planned Enhancements

- Redis caching
- background job queues
- email retry service
- advanced analytics
- AI lead scoring
- microservices architecture
- CI/CD pipelines
- cloud deployment
- mobile application

Conclusion

The project demonstrates:

- full stack development
- scalable architecture
- secure authentication
- enterprise RBAC
- modern UI/UX
- production-level engineering practices